

Stabilization of Stochastic Uncertain Complex-Valued Delayed Networks via Aperiodically Intermittent Nonlinear Control

Pengfei Wang, Biguang Zhang, and Huan Su 

Abstract—This paper concerns the stabilization problem of stochastic uncertain complex-valued delayed networks (SUCVDNs) via aperiodically intermittent nonlinear control. It is worth noting that the intermittent control is aperiodic and nonlinear. As a special case, when each control width tends to zero, intermittent control becomes impulsive control. Hence, the stabilization problem of SUCVDNs via impulsive control is also studied. The main method is the combination of the Lyapunov method and Kirchhoff's matrix tree theorem in graph theory, which does not require us to solve any linear matrix inequalities. Different from previous results on complex-valued system, we consider the complex-valued system on complex space directly by using complex version Itô's formula and (conjugate) $\mathbb{R}$ derivative. Then we give some sufficient criteria, which have a close relationship with the control interval and the topological structure of the considered network. The main results are applied to a class of stochastic complex-valued coupled oscillators, and numerical examples are also presented to show the effectiveness of the control strategies.

Index Terms—Aperiodically intermittent nonlinear control, (conjugate) $\mathbb{R}$ derivative, impulsive control, stabilization, stochastic complex-valued networks.

I. INTRODUCTION

OVER the past decades, much attention has been paid to the stability of coupled networks due to their immense potential applications in various fields, such as biology, neural networks, and so on [1]–[7]. Moreover, in various systems, time delay is a very common phenomenon. For instance, most of the sensors have time delay because of the process of their measurements. Since time delay can cause poor performance or even instability, it is necessary to take time delay into account when studying the stability of coupled networks. On

the other hand, due to the inaccuracies and/or changes in the practical environment of models, we cannot avoid parameter uncertainties in coupled networks [8]–[11]. And in our real lives, systems are commonly subject to the influence of various factors such as environmental noise [12]–[14]. In this case, it is of great importance to study the stability for stochastic uncertain delayed networks.

Recently, complex-valued coupled networks have received increasing interest since many practical problems can be better solved by complex-valued systems. For example, the XOR problem cannot be solved by a single real-valued neuron and only one complex-valued neuron is sufficient [15]. In wave-related adaptive process, we often obtain excellent performance with learning or self-organization based on the complex-valued neural networks compared with real-valued neural networks [16]. When modeling wind profile [17], the fully complex approach (i.e., simultaneously speed and direction as a complex vector) has major advantages over the dual-univariate approach (i.e., speed and direction as independent processes). And until now, some stability criteria have been given for various complex-valued networks. For example, in [18], by using Matrix measure method, the global exponential stability of complex-valued recurrent neural networks was considered. Hu and Wang [19] and Wang *et al.* [20] studied the stability of complex-valued (memristive) recurrent neural networks by means of M -matrix and Lyapunov function. In [21], adaptive exponential synchronization of complex-valued Cohen–Grossberg neural networks with known and unknown parameters was investigated. For more engineering applications, we refer the reader to [22]–[24] and the references therein. It should be pointed out that the aforementioned references dealt with the complex-valued systems by decomposing them to two real-valued systems. However, this approach demands us to find an explicit separation of the complex-valued function into its real imaginary parts. But in most cases, it is difficult to express this separation in an analytical form. Moreover, compared with the original complex-valued systems, the double dimension of the real-valued systems leads to significant analysis difficulties. Hence, it is more effective to study complex-valued systems directly in the complex domain. However, the derivative of many functions with complex variables [such as $f(x) = x\bar{x}$ where $x \in \mathbb{C}$] does not exist since they are not holomorphic, which leads to great studying difficulty. In this paper, by introducing complex version Itô's formula [25] and (conjugate) $\mathbb{R}$

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derivative [26], we can overcome above-mentioned difficulty and avoid separating systems' real and imaginary parts.

In many practical systems, it is difficult to realize self-stabilization, so the stabilization control problem tends to be a hot topic in both theoretical researches and practical applications. Particularly, intermittent control [27] has drawn noticeable attention since it reduces control cost and the amount of the information sent on networks. Usually, intermittent control is classified into periodically intermittent control [28]–[30] and aperiodically intermittent control [31]–[35]. Actually, aperiodically intermittent control is more general and takes periodically intermittent control as a special case. However, above results on aperiodically intermittent control all considered linear feedback controller and studied real-valued systems. Whereas, nonlinear case is more general. Considering the practical meaning of complex-valued systems, in this paper, we will consider the stabilization problem of stochastic uncertain complex-valued delayed networks (SUCVDNs) via aperiodically intermittent nonlinear control, which has not been studied according to our survey. What is more, when control width tends to zero, the intermittent control becomes impulsive control, which was not considered in [27]–[35]. Due to the increasing importance on impulsive control [36]–[38], in this paper, we will also present a detailed discussion on the stabilization of SUCVDNs via impulsive control.

Motivated by above discussions, in this paper, we will investigate the stabilization problem of SUCVDNs via aperiodically intermittent nonlinear control. As we all know, the Lyapunov method is an effective method to study the stability of systems. However, the high dimension for coupled networks, especially for complex-valued networks, makes it difficult to directly construct a suitable Lyapunov function. Nowadays, many researchers have devoted themselves to studying the stability and stabilization problem of coupled networks via the Lyapunov method and linear matrix inequality techniques [20]–[23], [39]–[41] or Kronecker product method [42], [43]. Li and Shuai [44] provided another effective method to explore the stability of high-dimensional coupled networks, which is the combination of the Lyapunov method and graph theory. Hence, inspired by [44], we will study the stabilization problem of SUCVDNs via aperiodically intermittent nonlinear control by combining the Lyapunov method and graph theory. Moreover, the impulsive control that can be seen as a special case of intermittent control when control width tends to zero is also considered. The novelties of this paper are summarized as below.

- 1) By introducing complex version Itô's formula and (conjugate) $\mathbb{R}$ derivative, we study SUCVDNs directly without separating their real and image parts.
- 2) The intermittent control we consider is aperiodic and nonlinear, which extends the previous results to some extent. Specially, the impulsive control is studied as well.
- 3) Intermittent control is applied to stabilizing complex-valued networks for the first time.
- 4) The theoretical results have a close relationship with the control interval and the topological structure of the considered networks.

The rest of this paper is organized as follows. We give some preliminaries and model descriptions in Section II. In Section III, some stability criteria of SUCVDNs are derived. An application to stochastic complex-valued coupled oscillators is given in Section IV. In Section V, numerical examples are presented to demonstrate the effectiveness of the theoretical results. Some conclusions are presented in Section VI.

II. PRELIMINARIES AND MODEL DESCRIPTIONS

Throughout this paper, let $\mathbb{C}^n$ and $\mathbb{C}^{m \times n}$ denote n -dimensional complex vectors and the set of $m \times n$ complex matrices, respectively. Let $\mathcal{N} = \{1, 2, \dots, N\}$, $\mathbb{N}_+ = \{1, 2, \dots\}$, $\mathbb{N} = \{0, 1, 2, \dots\}$, $\mathbb{R} = (-\infty, +\infty)$, and $\mathbb{R}_+ = [0, +\infty)$. Then, for any complex number (or complex vector) x , x^R , and x^I denote its real and imaginary parts, respectively, and x^T , $\bar{x}$ is the transpose and conjugated of x , respectively. The norm of any complex vector is $\|x\| = \sqrt{x^T \bar{x}}$. Denote $\lambda_{\max}(\cdot)$ and $\lambda_{\min}(\cdot)$ as the maximal and minimal eigenvalues of a Hermite matrix, respectively. The norm of matrix A is $\|A\| = \sqrt{\lambda_{\max}(A^T A)}$. Denote $\mathbf{i}$ by the imaginary unit, i.e., $\mathbf{i} = \sqrt{-1}$. Let I_n represent n -dimensional identity matrix. Moreover, let $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \geq 0}, \mathbb{P})$ be a complete probability space with filtration $\{\mathcal{F}_t\}_{t \geq 0}$ satisfying the usual conditions (i.e., the filtration contains all $\mathbb{P}$ -null sets and is right continuous). Denote by $L_{\mathcal{F}_0}^2([-\tau, 0]; \mathbb{C}^n)$ the family of all $\mathcal{F}_0$ -measurable $\mathcal{C}([-\tau, 0]; \mathbb{C}^n)$ -valued random variable $\xi = \{\xi(s) : -\tau \leq s \leq 0\}$ such that $\sup_{-\tau \leq s \leq 0} \mathbb{E}\|\xi(s)\|^2 < \infty$, where $\mathbb{E}\{\cdot\}$ stands for mathematical expectation operator with respect to the given probability measure $\mathbb{P}$.

Then, some basic concepts on graph theory are given as follows: a digraph $\mathcal{G} = (\mathcal{N}, \mathcal{E})$ contains a set of vertices $\mathcal{N}$ and a set $\mathcal{E}$ of arcs (i, h) leading from initial vertex i to terminal vertex h , satisfying $(i, i) \notin \mathcal{E}$ for all $i \in \mathcal{N}$. A digraph $\mathcal{G}$ is weighted if each arc (h, i) is assigned a positive weight a_{ih} . Let $A = (a_{ih})_{N \times N}$ represent the weight matrix of $\mathcal{G}$, where $a_{ih} > 0$ if and only if there exists an arc (h, i) in $\mathcal{G}$. A digraph is strongly connected, if for any pair of distinct vertices, there exists a directed path from one to another. Denote digraph $\mathcal{G}$ with weight matrix A as $(\mathcal{G}, A)$. The Laplacian matrix of $\mathcal{G}$ is denoted by $L(\mathcal{G}) = (\chi_{ih})_{N \times N}$, where

$$\chi_{ih} = \begin{cases} \sum_{r \neq i} a_{ir}, & i = h \\ -a_{ih}, & i \neq h. \end{cases}$$

For other details on graph theory, we refer the reader to [45].

We describe an SUCVDN consisting of N nodes on a digraph $\mathcal{G}$ with N vertices as follows:

$$\begin{aligned} dx_i(t) = & \left[(A_i + \Delta A_i(t))x_i(t) + (B_i + \Delta B_i(t))x_i(t - \tau(t)) \right. \\ & + f_i(t, x_i(t), x_i(t - \tau(t))) \\ & + \sum_{h=1}^N H_{ih}(x_i(t - \tau(t)), x_h(t - \tau(t))) + u_i(t) \Big] dt \\ & + h_i(x_i(t), x_i(t - \tau(t)))dB(t), \quad i \in \mathcal{N} \end{aligned} \quad (1)$$

where $x_i(t) \in \mathbb{C}^n$, $n \in \mathbb{N}_+$, $A_i, B_i \in \mathbb{C}^{n \times n}$, $\Delta A_i(t), \Delta B_i(t) \in \mathbb{C}^{n \times n}$ are parameter perturbation matrices bounded by: $\|\Delta A_i(t)\|^2 \leq \gamma_{i1}^2$, $\|\Delta B_i(t)\|^2 \leq \gamma_{i2}^2$ for all $t \in \mathbb{R}_+$,

function $f_i(\cdot, \cdot, \cdot) : \mathbb{R}_+ \times \mathbb{C}^n \times \mathbb{C}^n \rightarrow \mathbb{C}^n$ is continuous, $h_i(\cdot, \cdot) : \mathbb{C}^n \times \mathbb{C}^n \rightarrow \mathbb{C}^n$ is a disturbance strength function, $H_{ih}(\cdot, \cdot) : \mathbb{C}^n \times \mathbb{C}^n \rightarrow \mathbb{C}^n$ represents the influence of vertex h on vertex i and $H_{ih} \equiv 0$ if there exists no arc from h to i in $\mathcal{G}$, $\tau(\cdot)$ is the time-varying delay bounded by known constant, i.e., $0 < \tau(\cdot) \leq \tau$, $B(\cdot)$ is a one-dimensional (1-D) Brownian motion defined on the complete probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \geq 0}, \mathbb{P})$, $u_i(t)$ is the aperiodically nonlinear intermittent control defined as

$$u_i(t) = \begin{cases} K_i(x_i(t)), & t_m \leq t \leq s_m \\ 0, & s_m < t < t_{m+1} \end{cases} \quad m \in \mathbb{N}.$$

Here, $t_0 = 0$, $K_i(\cdot) : \mathbb{C}^n \rightarrow \mathbb{C}^n$ is the control gain function and satisfies $K_i(0) = 0$. For any time span $[t_m, t_{m+1}]$, $[t_m, s_m]$ is the m th control time (work time), and $s_m - t_m$ is the m th control width; while (s_m, t_{m+1}) is the m th rest time, and $t_{m+1} - s_m$ is the m th rest width.

Remark 1: When $t_m \equiv mT$, $s_m \equiv (m + \theta)T$, where $0 < \theta < 1$, aperiodically intermittent control degenerates into periodically intermittent control, which was studied in [27]–[30]. Moreover, if $K_i(x_i(t)) = -k_i x_i(t)$, where k_i is a positive constant, then aperiodically intermittent nonlinear control becomes linear type, which was considered in [31]–[35]. Hence, the control considered in this paper is more general. Obviously, intermittent control turns into impulse signals when $s_m \rightarrow t_m$, which has not been considered in existing results on aperiodically intermittent control.

Remark 2: It should be pointed that in most literatures, the coupling form was considered as $H_{ih}(x_h)$, which is independent of x_i , see [8], [9], [18], [23], [27]–[30], [38]. While in many institutions, the coupling form H_{ih} has a close relationship with both x_i and x_h . Take Kuramoto oscillators [1] and multipath population model [2] respectively as examples, the coupling forms are $H_{ih}(x_i, x_h) = \sin(x_i - x_h)$ and $H_{ih}(x_i, x_h) = x_h - \alpha_{ih} x_i$, respectively. Moreover, for time delay $\tau(t)$, we do not require it to be differentiable. In [14], [27], [28], and [30], it is necessary for $\tau(t)$ to satisfy $\tau(t) \leq \tau$ and $\dot{\tau} \leq \tilde{\tau} < \infty$. Hence, the coupling form and time delay considered in this paper are more general. However, this general coupling form restrict us to using pinning control (i.e., controllers are added to only a small fraction of nodes). In general, networks with coupling form like $H_{ih}(x_i, x_h) = x_i - x_h$ can be well investigated by pinning control and Kronecker product method [42], [43].

Let $\phi(s) = (\phi_1^T(s), \dots, \phi_N^T(s))^T \in L_{\mathcal{F}_0}^2([-\tau, 0]; \mathbb{C}^{nN})$ be the initial conditions. As usual, we always assume that the functions $f_i(\cdot, \cdot, \cdot)$, $H_{ih}(\cdot, \cdot)$, $h_i(\cdot, \cdot)$ ($i, h \in \mathcal{N}$) and $K_i(\cdot)$ satisfy all necessary conditions to ensure that system (1) has a unique solution on the entire $t \geq -\tau$, which is denoted by $x(t, \phi)$. For simplicity, we write $x(t, \phi) = x(t) = (x_1^T(t), \dots, x_N^T(t))^T \in \mathbb{C}^{nN}$. Moreover, we also assume that $f_i(t, 0, 0) = H_{ih}(0, 0) = h_i(0, 0) = 0$ ($i, h \in \mathcal{N}$). So system (1) admits a trivial solution $x(t) \equiv 0$ corresponding to the initial condition $\phi(s) = 0$ ($s \in [-\tau, 0]$).

The following definitions and lemmas will be used in deriving the main results of this paper.

Definition 1 [14]: System (1) is said to be globally exponentially stable in mean square if

there exist positive constants ι , γ such that for any initial conditions $\phi(s) \in L_{\mathcal{F}_0}^2([-\tau, 0]; \mathbb{C}^{nN})$, $\mathbb{E}\|x(t)\|^2 \leq \sup_{-\tau \leq s \leq 0} \mathbb{E}\|\phi(s)\|^2 \gamma \exp(-\iota t)$ holds for $t \geq 0$.

Definition 2 [13], [25]: Consider an n -dimensional complex-valued stochastic differential equation

$$dx(t) = f(t, x(t), x(t - \tau(t)))dt + \sigma(t, x(t), x(t - \tau(t)))dB(t)$$

where $x(t) = (x_1(t), x_2(t), \dots, x_n(t))^T \in \mathbb{C}^n$. For $V(t, x) : \mathbb{R}_+ \times \mathbb{C}^n \rightarrow \mathbb{R}_+$ [in fact, we can write $V(t, x) = V(t, x, \bar{x})$], define the $\mathbb{R}$ derivative of V as

$$\left. \frac{\partial V(t, x)}{\partial x} \right|_{\bar{x}=\text{con}} = \left(\frac{\partial V(t, x)}{\partial x_1}, \dots, \frac{\partial V(t, x)}{\partial x_n} \right) \Big|_{\bar{x}=\text{con}}$$

and define the conjugate $\mathbb{R}$ derivative of V as

$$\left. \frac{\partial V(t, x)}{\partial \bar{x}} \right|_{x=\text{con}} = \left(\frac{\partial V(t, x)}{\partial \bar{x}_1}, \dots, \frac{\partial V(t, x)}{\partial \bar{x}_n} \right) \Big|_{x=\text{con}}$$

where con represents constant. Let $\mathcal{C}^{1,2}(\mathbb{R}_+ \times \mathbb{C}^n; \mathbb{R}_+)$ denote the family of all nonnegative functions $V(t, x)$ on $\mathbb{R}_+ \times \mathbb{C}^n$, which are once continuously differentiable in t and twice differentiable in x and $\bar{x}$. Then for $V \in \mathcal{C}^{1,2}(\mathbb{R}_+ \times \mathbb{C}^n; \mathbb{R}_+)$, the complex version of Itô's formula takes the following form:

$$\begin{aligned} dV(t, x) &= \frac{\partial V(t, x)}{\partial t} dt + \frac{\partial V(t, x)}{\partial x} dx + \frac{\partial V(t, x)}{\partial \bar{x}} d\bar{x} \\ &\quad + \frac{1}{2} \sum_{p,q=1}^n \frac{\partial^2 V(t, x)}{\partial x_p \partial x_q} dx_p dx_q + \sum_{p,q=1}^n \frac{\partial^2 V(t, x)}{\partial x_p \partial \bar{x}_q} dx_p d\bar{x}_q \\ &\quad + \frac{1}{2} \sum_{p,q=1}^n \frac{\partial^2 V(t, x)}{\partial \bar{x}_p \partial \bar{x}_q} d\bar{x}_p d\bar{x}_q \\ &= \mathcal{L}V(t, x)dt + \left[\frac{\partial V(t, x)}{\partial x} \sigma(t) + \frac{\partial V(t, x)}{\partial \bar{x}} \bar{\sigma}(t) \right] dB(t) \end{aligned}$$

in which

$$\begin{aligned} f(t) &= f(t, x(t), x(t - \tau(t))), \sigma(t) = \sigma(t, x(t), x(t - \tau(t))) \\ \frac{\partial^2 V(t, x)}{\partial x^2} &= \left(\frac{\partial^2 V(t, x)}{\partial x_p \partial x_q} \right)_{n \times n} \\ \frac{\partial^2 V(t, x)}{\partial x \partial \bar{x}} &= \left(\frac{\partial^2 V(t, x)}{\partial x_p \partial \bar{x}_q} \right)_{n \times n} \\ \frac{\partial^2 V(t, x)}{\partial \bar{x}^2} &= \left(\frac{\partial^2 V(t, x)}{\partial \bar{x}_p \partial \bar{x}_q} \right)_{n \times n} \\ dB(t)dB(t) &= dt, \quad dB(t)dt = dt dB(t) = dt dt = 0 \end{aligned}$$

and operator $\mathcal{L}V(t, x)$ is defined as

$$\begin{aligned} \mathcal{L}V(t, x) &= \frac{\partial V(t, x)}{\partial t} + \frac{\partial V(t, x)}{\partial x} f(t) + \frac{\partial V(t, x)}{\partial \bar{x}} \bar{f}(t) \\ &\quad + \frac{1}{2} \sigma^T(t) \frac{\partial^2 V(t, x)}{\partial x^2} \sigma(t) + \sigma^T(t) \frac{\partial^2 V(t, x)}{\partial x \partial \bar{x}} \bar{\sigma}(t) \\ &\quad + \frac{1}{2} \bar{\sigma}^T(t) \frac{\partial^2 V(t, x)}{\partial \bar{x}^2} \bar{\sigma}(t). \end{aligned}$$

In Definition 2, $(\partial/\partial x)$ is called $\mathbb{R}$ -derivative and $(\partial/\partial \bar{x})$ is called conjugate $\mathbb{R}$ -derivative [26]. In fact, the complex version of Itô's formula (for 1-D complex-valued system) was mentioned in [25]. Definition 2 can be analogized according [13].

Lemma 1 [36]: For any $X, Y \in \mathbb{C}^n$ and constant $\zeta > 0$, the following inequality holds:

$$X^T \bar{Y} + \bar{X}^T Y \leq \zeta \|X\|^2 + \zeta^{-1} \|Y\|^2.$$

Lemma 2 [31]: Let $z : [t_0 - \tau, \infty) \rightarrow [0, \infty)$ be a continuous function such that $\dot{z}(t) \leq -\alpha z(t) + \gamma \max\{z_t\}$ is satisfied for $t \geq t_0$. If $\alpha > \gamma > 0$, then $z(t) \leq \max\{z_{t_0}\} \exp(-\epsilon(t - t_0))$, $t \geq t_0$, where $\max\{z_{t_0}\} = \sup_{t_0 - \tau \leq \xi \leq t_0} z(\xi)$ and $\epsilon > 0$ is the unique real root of equation $\alpha - \epsilon - \gamma \exp(\epsilon \tau) = 0$.

Lemma 3 [31]: Let $z : [t_0 - \tau, \infty) \rightarrow [0, \infty)$ be a continuous function such that $\dot{z}(t) \leq \delta z(t) + \gamma \max\{z_t\}$ is satisfied for $t \geq t_0$. If $\delta > 0$, $\gamma > 0$, then $z(t) \leq \max\{z_{t_0}\} \exp((\delta + \gamma)(t - t_0))$, $t \geq t_0$, where $\max\{z_{t_0}\} = \sup_{t_0 - \tau \leq \xi \leq t_0} z(\xi)$.

Lemma 4 [37]: Let $0 \leq \tau_1(t), \tau_2(t), \dots, \tau_m(t) \leq \tau = \max_{m+1} \{\tau_1, \tau_2, \dots, \tau_m\}$, $F(t, z, \tilde{z}_1, \dots, \tilde{z}_m) : \mathbb{R}_+ \times \underbrace{\mathbb{R} \times \dots \times \mathbb{R}}_{m+1} \rightarrow \mathbb{R}$ be nondecreasing in $\tilde{z}_i$ for each fixed $(t, z, \tilde{z}_1, \dots, \tilde{z}_{i-1}, \tilde{z}_{i+1}, \dots, \tilde{z}_m)$, $i = 1, 2, \dots, m$, and $I_k(z) : \mathbb{R} \rightarrow \mathbb{R}$ be nondecreasing in z . Suppose that $z(t)$ and $v(t) \in C([- \tau, 0]; \mathbb{R})$ satisfy

$$\begin{cases} D^+ z(t) \leq F(t, z(t), z(t - \tau_1(t)), \dots, z(t - \tau_m(t))), t \geq 0 \\ z(t_k^+) \leq I_k(z(t_k^-)), k \in \mathbb{N}_+ \\ D^+ v(t) \leq F(t, v(t), v(t - \tau_1(t)), \dots, v(t - \tau_m(t))), t \geq 0 \\ v(t_k^+) \leq I_k(v(t_k^-)), k \in \mathbb{N}_+ \end{cases}$$

where the right and upper Dini's derivative $D^+ z(t)$ is defined as $D^+ z(t) = \overline{\lim}_{q \rightarrow 0^+} [(z(t + q) - z(t))/q]$, where $q \rightarrow 0^+$ means that q approaches zero from the right-hand side. Then $z(t) \leq v(t)$ for $-\tau \leq t \leq 0$ implies that $z(t) \leq v(t)$ for $t \geq 0$.

III. MAIN RESULTS

In this section, we give several exponential stabilization criteria for system (1) via aperiodically intermittent nonlinear control. In Section III-A, a Lyapunov-type theorem which is related to vertex-Lyapunov functions (see Definition 3) is presented based on the Lyapunov method and graph theory. In Section III-B, we explore another easily verified exponential stabilization criterion regarding to coefficients of system (1). Finally, a special case for intermittent control that control time tends to zero is discussed in Section III-C.

A. Lyapunov-Type Theorem

First, we introduce the following definition and assumption.

Definition 3: Functions $V_i(t, x_i) \in \mathcal{C}^{1,2}(\mathbb{R}_+ \times \mathbb{C}^n; \mathbb{R}_+)$, $i \in \mathcal{N}$ are called vertex-Lyapunov functions for system (1) if the following conditions are satisfied.

A1: For any $i \in \mathcal{N}$, there exist positive constants ϑ_i, μ_i such that

$$\vartheta_i \|x_i\|^2 \leq V_i(t, x_i) \leq \mu_i \|x_i\|^2, \quad t \geq 0. \quad (2)$$

A2: There exist nonnegative constants $\xi_i, \sigma_i, \alpha_i$, and a_{ih} and functions F_{ih} ($i, h \in \mathcal{N}$), such that

$$\begin{aligned} \mathcal{L}V_i(t, x_i(t)) &\leq -\sigma_i V_i(t, x_i(t)) + \xi_i V_i(t - \tau(t), x_i(t - \tau(t))) \\ &\quad + \sum_{h=1}^N a_{ih} F_{ih}(x_i(t - \tau(t)), x_h(t - \tau(t))) \end{aligned} \quad (3)$$

for $t \in [t_m, s_m]$ and

$$\begin{aligned} \mathcal{L}V_i(t, x_i(t)) &\leq \alpha_i V_i(t, x_i(t)) + \xi_i V_i(t - \tau(t), x_i(t - \tau(t))) \\ &\quad + \sum_{h=1}^N a_{ih} F_{ih}(x_i(t - \tau(t)), x_h(t - \tau(t))) \end{aligned} \quad (4)$$

for $t \in (s_m, t_{m+1})$, $m \in \mathbb{N}$.

A3: Along each directed cycle $\mathcal{C}$ of the weighted digraph $(\mathcal{G}, (a_{ih})_{N \times N})$, it holds that

$$\sum_{(r,s) \in \mathcal{C}} F_{sr}(x_s, x_r) \leq 0. \quad (5)$$

Assumption 1 [31]: For the aperiodically intermittent control, there exist two positive scalars $0 < \theta < T < +\infty$, such that, for $m \in \mathbb{N}$

$$\begin{cases} \inf_m (s_m - t_m) = \theta \\ \sup_m (t_{m+1} - s_m) = T. \end{cases}$$

Theorem 1: Let Assumption 1 hold and $(\mathcal{G}, A)$ ($A = (a_{ih})_{N \times N}$) be strongly connected. Suppose that system (1) admits vertex-Lyapunov functions $V_i(t, x_i)$, $i \in \mathcal{N}$, and the following inequalities hold:

$$(\theta - \tau)\lambda - (T - \theta)(\alpha + \xi) > 0, \quad \sigma > \xi \quad (6)$$

where $\alpha = \max_{i \in \mathcal{N}} \{\alpha_i\}$, $\sigma = \min_{i \in \mathcal{N}} \{\sigma_i\}$, $\xi = \max_{i \in \mathcal{N}} \{\xi_i\}$, $\lambda > 0$ is the unique real root of $\sigma - \lambda - \xi \exp(\lambda \tau) = 0$. Then the trivial solution of system (1) is globally exponentially stable in mean square.

Proof: Define

$$V(t, x) = \sum_{i=1}^N c_i V_i(t, x_i)$$

where c_i is the cofactor of the i th diagonal element of $L(\mathcal{G})$. Since $(\mathcal{G}, A)$ is strongly connected, we have $c_i > 0$ ($i \in \mathcal{N}$). Then, by virtue of (2), we can obtain

$$\begin{aligned} V(t, x) &= \sum_{i=1}^N c_i V_i(t, x_i) \geq \sum_{i=1}^N c_i \vartheta_i \|x_i\|^2 \\ &\geq \min_{i \in \mathcal{N}} \{c_i \vartheta_i\} \sum_{i=1}^N \|x_i\|^2 = \vartheta \|x\|^2 \end{aligned}$$

where $\vartheta = \min_{i \in \mathcal{N}} \{c_i \vartheta_i\} > 0$. Similarly, we have $V(t, x) \leq \mu \|x\|^2$, where $\mu = \max_{i \in \mathcal{N}} \{c_i \mu_i\} > 0$. Hence,

$$\vartheta \|x\|^2 \leq V(t, x) \leq \mu \|x\|^2. \quad (7)$$

When $t \in [t_m, s_m]$, by virtue of (3), we can arrive at

$$\begin{aligned} \mathcal{L}V(t, x(t)) &= \sum_{i=1}^N c_i \mathcal{L}V_i(t, x_i(t)) \\ &\leq - \sum_{i=1}^N c_i \sigma_i V_i(t, x_i(t)) \\ &\quad + \sum_{i=1}^N c_i \xi_i V_i(t - \tau(t), x_i(t - \tau(t))) \\ &\quad + \sum_{i,h=1}^N c_i a_{ih} F_{ih}(x_i(t - \tau(t)), x_h(t - \tau(t))). \end{aligned} \quad (8)$$

By Kirchhoff's matrix tree theorem [44, Th. 2.2] and (5), we have

$$\sum_{i,h=1}^N c_i a_{ih} F_{ih}(x_i, x_h) = \sum_{\mathcal{Q} \in \mathbb{Q}} W(\mathcal{Q}) \sum_{(r,s) \in \mathcal{E}(\mathcal{C}_{\mathcal{Q}})} F_{sr}(x_s, x_r) \leq 0 \quad (9)$$

where $\mathbb{Q}$ is the set of all spanning unicyclic graphs of $(\mathcal{G}, A)$, $W(\mathcal{Q})$ is the weight of $\mathcal{Q}$ and $\mathcal{C}_{\mathcal{Q}}$ denotes the directed cycle of $\mathcal{Q}$. By virtue of (8) and (9), we have

$$\mathcal{L}V(t, x(t)) \leq -\sigma V(t, x(t)) + \xi V(t - \tau(t), x(t - \tau(t))). \quad (10)$$

Utilizing Itô's formula, we have

$$D^+ \mathbb{E}V(t, x(t)) \leq -\sigma \mathbb{E}V(t, x(t)) + \xi \mathbb{E}V(t - \tau(t), x(t - \tau(t))). \quad (11)$$

By Lemma 2 and (11), we can get that

$$\mathbb{E}V(t, x(t)) \leq \sup_{t_m - \tau \leq s \leq t_m} \mathbb{E}V(s, x(s)) \exp(-\lambda(t - t_0)) \quad (12)$$

where $\lambda > 0$ is the unique real root of equation $\sigma - \lambda - \xi \exp(\lambda\tau) = 0$.

On the other hand, when $t \in (s_m, t_{m+1})$, using (4), we can similarly obtain

$$\begin{aligned} \mathcal{L}V(t, x(t)) &= \sum_{i=1}^N c_i \mathcal{L}V_i(t, x_i(t)) \\ &\leq \sum_{i=1}^N c_i \alpha_i V_i(t, x_i(t)) \\ &\quad + \sum_{i=1}^N c_i \xi_i V_i(t - \tau(t), x_i(t - \tau(t))) \\ &\quad + \sum_{i,h=1}^N c_i a_{ih} F_{ih}(x_i(t - \tau(t)), x_h(t - \tau(t))) \\ &\leq \alpha V(t, x(t)) + \xi V(t - \tau(t), x(t - \tau(t))). \end{aligned}$$

Then we have

$$D^+ \mathbb{E}V(t, x(t)) \leq \alpha \mathbb{E}V(t, x(t)) + \xi \mathbb{E}V(t - \tau(t), x(t - \tau(t))). \quad (13)$$

According to Lemma 3 and (13), we can derive

$$\mathbb{E}V(t, x(t)) \leq \sup_{s_m - \tau \leq s \leq s_m} \mathbb{E}V(s, x(s)) \exp((\alpha + \xi)(t - t_0)). \quad (14)$$

When $m = 0$, we can easily get the following inequalities. For $t \in [0, s_0]$, employing inequality (12), we can obtain

$$\mathbb{E}V(t, x(t)) \leq \sup_{-\tau \leq s \leq 0} \mathbb{E}V(s, x(s)) \exp(-\lambda t).$$

For $t \in (s_0, t_1)$, according to inequality (14), we can get

$$\begin{aligned} \mathbb{E}V(t, x(t)) &\leq \sup_{s_0 - \tau \leq s \leq s_0} \mathbb{E}V(s, x(s)) \exp((\alpha + \xi)(t - s_0)) \\ &\leq \sup_{-\tau \leq s \leq 0} \mathbb{E}V(s, x(s)) \\ &\quad \times \exp(-\lambda(s_0 - \tau) + (\alpha + \xi)(t - s_0)). \end{aligned}$$

When $m = p$, we assume the following inequalities hold:

$$\begin{aligned} \mathbb{E}V(t, x(t)) &\leq \sup_{-\tau \leq s \leq 0} \mathbb{E}V(s, x(s)) \\ &\quad \times \exp(-\lambda(t - t_p) - p\lambda(\theta - \tau) \\ &\quad + p(\alpha + \xi)(T - \theta)), t \in (t_p, s_p] \end{aligned} \quad (15)$$

and

$$\begin{aligned} \mathbb{E}V(t, x(t)) &\leq \sup_{-\tau \leq s \leq 0} \mathbb{E}V(s, x(s)) \\ &\quad \times \exp((\alpha + \xi)(t - s_p) + p(\alpha + \xi)(T - \theta) \\ &\quad - (p + 1)\lambda(\theta - \tau)), t \in (s_p, t_{p+1}]. \end{aligned} \quad (16)$$

Then we discuss the case for $m = p + 1$. For $t \in [t_{p+1}, s_{p+1}]$, combining Lemma 2 and (15), we obtain

$$\begin{aligned} \mathbb{E}V(t, x(t)) &\leq \sup_{t_{p+1} - \tau \leq s \leq t_{p+1}} \mathbb{E}V(s, x(s)) \exp(-\lambda(t - t_{p+1})) \\ &\leq \sup_{-\tau \leq s \leq 0} \mathbb{E}V(s, x(s)) \\ &\quad \times \exp(-\lambda(t - t_{p+1}) - (p + 1)\lambda(\theta - \tau) \\ &\quad + (p + 1)(\alpha + \xi)(T - \theta)). \end{aligned}$$

For $t \in (s_{p+1}, t_{p+2})$, combining Lemma 3 with (16), we derive that

$$\begin{aligned} \mathbb{E}V(t, x(t)) &\leq \sup_{s_{p+1} - \tau \leq s \leq s_{p+1}} \mathbb{E}V(s, x(s)) \exp((\alpha + \xi)(t - s_{p+1})) \\ &\leq \sup_{-\tau \leq s \leq 0} \mathbb{E}V(s, x(s)) \\ &\quad \times \exp((\alpha + \xi)(t - s_{p+1}) + (p + 1)(\alpha + \xi)(T - \theta) \\ &\quad - (p + 2)\lambda(\theta - \tau)). \end{aligned}$$

Hence, utilizing mathematical induction, the case of $m = p$ holds for $t \in [0, +\infty)$.

For $t \in [t_m, t_{m+1})$, we can get $-m \leq -(t/T) + 1$. For $t \in [t_m, s_m]$, by the fact that $(\theta - \tau)\lambda - (T - \theta)(\alpha + \xi) > 0$, we have

$$\begin{aligned} \mathbb{E}V(t, x(t)) &\leq \sup_{-\tau \leq s \leq 0} \mathbb{E}V(s, x(s)) \\ &\quad \times \exp(-\lambda(t - t_m) - m\lambda(\theta - \tau) \\ &\quad + m(\alpha + \xi)(T - \theta)) \\ &\leq \sup_{-\tau \leq s \leq 0} \mathbb{E}V(s, x(s)) \\ &\quad \times \exp(-m\lambda(\theta - \tau) + m(\alpha + \xi)(T - \theta)) \\ &\leq \sup_{-\tau \leq s \leq 0} \mathbb{E}\tilde{M}V(s, x(s)) \exp(-\lambda t) \end{aligned}$$

where $\tilde{M} = \exp((\theta - \tau)\lambda - (T - \theta)(\alpha + \xi)) > 0$, $\iota = (1/T)[(\theta - \tau)\lambda - (T - \theta)(\alpha + \xi)] > 0$. For $t \in (s_m, t_{m+1})$

$$\begin{aligned} \mathbb{E}V(t, x(t)) &\leq \sup_{-\tau \leq s \leq 0} \mathbb{E}V(s, x(s)) \\ &\quad \times \exp((\alpha + \xi)(t - s_m) + m(\alpha + \xi)(T - \theta) \\ &\quad - (m + 1)\lambda(\theta - \tau)) \\ &\leq \sup_{-\tau \leq s \leq 0} \mathbb{E}V(s, x(s)) \\ &\quad \times \exp(((\alpha + \xi)T - m((\theta - \tau)\lambda - (T - \theta)(\alpha + \xi)))) \\ &\leq \sup_{-\tau \leq s \leq 0} \mathbb{E}V(s, x(s))M \exp(-\iota t) \end{aligned}$$

where $M = \exp((\alpha + \xi)T + (\theta - \tau)\lambda - (T - \theta)(\alpha + \xi)) > \tilde{M} > 0$.

In summary, we arrive at the following inequality for $t \in [0, +\infty)$:

$$\mathbb{E}V(t, x(t)) \leq M \sup_{-\tau \leq s \leq 0} \mathbb{E}V(s, x(s)) \exp(\theta T \lambda (-\iota t)).$$

Then, combining with (7), it yields

$$\mathbb{E}\|x(t)\|^2 \leq \frac{1}{\vartheta} \mathbb{E}V(t, x(t)) \leq \frac{\mu M}{\vartheta} \sup_{-\tau \leq s \leq 0} \mathbb{E}\|\phi(s)\|^2 \exp(-\iota t).$$

Therefore, from Definition 1, we derive that system (1) is globally exponentially stable in mean square. The proof is completed. ■

Remark 3: Note that if we can find functions $\Xi_i(x_i)$ ($i \in \mathcal{N}$) such that

$$F_{ih}(x_i, x_h) \leq \Xi_h(x_h) - \Xi_i(x_i), \quad i, h \in \mathcal{N}$$

then condition A3 can be easily satisfied according to [44].

Remark 4: Recently, intermittent control has been widely applied to studying the synchronization and stability of complex networks [27]–[35]. These results all consider the complex network as a whole system, which cannot reflect the inner topological structure. Compared with these results, we treat each node as a subsystem and consider each subsystem first. Then we consider the whole network by means of the topological structure of considered networks. In detail, in Theorem 1, we construct Lyapunov function $V(t, x)$ as following:

$$V(t, x) = \sum_{i=1}^N c_i V_i(t, x_i)$$

where c_i is the cofactor of the i th diagonal element of $L(\mathcal{G})$ and $c_i > 0$ since digraph $(\mathcal{G}, A)$ is strongly connected. This means that $V(t, x)$ has a close relationship with topological properties of digraph $(\mathcal{G}, A)$.

Remark 5: Theorem 1 desires digraph $(\mathcal{G}, A)$ to be strongly connected. Actually, the strongly connected condition is indispensable. If the weighted digraph $(\mathcal{G}, A)$ is not strongly connected, only parts of vertices system can be stabilized, which implies that the whole controlled system may not be stable. We give an example as follows to illustrate the case.

Consider a weighted digraph $(\mathcal{G}, A)$ with four vertices which is not strongly connected, where

$$A = (a_{ih})_{4 \times 4} = \begin{bmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 \end{bmatrix}.$$

We can easily obtain that Laplacian matrix

$$L(\mathcal{G}) = (\chi_{ih})_{4 \times 4} = \begin{bmatrix} 5 & -1 & -2 & -2 \\ -1 & 3 & -2 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & -1 & 2 \end{bmatrix}.$$

By simple calculation, we have $c_1 = c_2 = c_4 = 0$ and $c_3 = 22$. It is obvious that, from the proof process of Theorem 1 and the construction of the Lyapunov function $V(t, x)$, we can get the stability of the third controlled vertex while we cannot guarantee that of the other vertices. So, from perspective of the whole coupled networks, the condition on strong connectedness of the networks is necessary and important.

In future work, will devoted to removing this restriction and consider a digraph with spanning tree [46], [47].

B. Coefficients-Type Theorem

In this part, we are going to find an easy-to-check criterion for global exponential stability in mean square of system (1), which is given with the coefficients of system (1). We first give the following assumptions on functions $f_i(\cdot, \cdot, \cdot)$, $H_{ih}(\cdot, \cdot)$, $h_i(\cdot, \cdot)$ and $K_i(\cdot)$.

Assumption 2: There exist constants $\Lambda_{ih} \geq 0$ and positive constants L_i , ϖ_i , and ς_i such that $f_i(t, x_i, y_i)$, $H_{ih}(x_i, y_h)$, and $h_i(x_i, y_i)$ satisfy

$$\begin{aligned} \|f_i(t, x_i, y_i)\|^2 &\leq L_i^2 (\|x_i\|^2 + \|y_i\|^2) \\ \|H_{ih}(x_i, x_h)\|^2 &\leq \Lambda_{ih}^2 (\|x_i\|^2 + \|x_h\|^2) \\ \|h_i(x_i, y_i)\|^2 &\leq \varpi_i \|x_i\|^2 + \varsigma_i \|y_i\|^2. \end{aligned} \quad (17)$$

Assumption 3: There are positive constants k_i , $i \in \mathcal{N}$, such that

$$\bar{x}_i^T K_i(x_i) + x_i^T \bar{K}_i(x_i) \leq -k_i \|x_i\|^2. \quad (18)$$

Theorem 2: Suppose that Assumptions 1–3 hold and that $(\mathcal{G}, (\Lambda_{ih})_{N \times N})$ is strongly connected. If the following inequalities are satisfied:

$$(\theta - \tau)\lambda - (T - \theta)(\alpha + \xi) > 0, \quad \sigma > \xi \quad (19)$$

where $\alpha = \max_{i \in \mathcal{N}} \{\lambda_{\max}(\Phi_i)\}$, $\lambda > 0$ is the real root of equation $\sigma - \lambda - \xi \exp(\lambda \tau) = 0$, $\sigma = \min_{i \in \mathcal{N}} \{\sigma_i\}$, $\xi = \max_{i \in \mathcal{N}} \{\xi_i\}$, $\sigma_i = \lambda_{\min}(k_i I_n - \Phi_i)$, $\xi_i = \|B_i\| + \gamma_{i2} + 2L_i + \sum_{h=1}^N \Lambda_{ih} + \varsigma_i$, $\Phi_i = \bar{A}_i + A_i^T + (\|B_i\| + 2\gamma_{i1} + \gamma_{i2} + 2L_i + \sum_{h=1}^N \Lambda_{ih} + \varpi_i)I_n$, then system (1) is globally exponentially stable in mean square.

Proof: Define

$$V_i(t, x_i) = x_i^T \bar{x}_i, \quad i \in \mathcal{N}.$$

It is obvious that condition A1 in Theorem 1 holds. Then according to Definition 2, we get that

$$\begin{aligned} \frac{\partial V_i(t, x_i)}{\partial x_i} &= \bar{x}_i^T, \quad \frac{\partial V_i(t, x_i)}{\partial \bar{x}_i} = x_i^T, \quad \frac{\partial^2 V_i(t, x_i)}{\partial x_i \partial \bar{x}_i} = I_n \\ \frac{\partial^2 V_i(t, x_i)}{\partial x_i^2} &= \frac{\partial^2 V_i(t, x_i)}{\partial \bar{x}_i^2} = 0. \end{aligned}$$

For $t \in [t_m, s_m]$, $m \in \mathbb{N}$, by Definition 2, the stochastic derivative of $V_i(t, x_i(t))$ along the trajectory of system (1) is calculated as follows:

$$\begin{aligned} \mathcal{L}V_i(t, x_i(t)) &= \bar{x}_i^T(t) \left[(A_i + \Delta A_i(t))x_i(t) + (B_i + \Delta B_i(t))x_i(t - \tau(t)) \right. \\ &\quad \left. + \bar{f}_i(t, x_i(t), x_i(t - \tau(t))) + K_i(x_i(t)) \right. \\ &\quad \left. + \sum_{h=1}^N H_{ih}(x_i(t - \tau(t)), x_h(t - \tau(t))) \right] \\ &\quad + x_i^T(t) \left[(\bar{A}_i + \overline{\Delta A}_i(t))\bar{x}_i(t) + (\bar{B}_i + \overline{\Delta B}_i(t))\bar{x}_i(t - \tau(t)) \right. \\ &\quad \left. + \bar{f}_i(t, x_i(t), x_i(t - \tau(t))) + \bar{K}_i(x_i(t)) \right. \\ &\quad \left. + \sum_{h=1}^N \bar{H}_{ih}(x_i(t - \tau(t)), x_h(t - \tau(t))) \right] \\ &+ h_i^T(x_i(t), x_i(t - \tau(t)))\bar{h}_i(x_i(t), x_i(t - \tau(t))) \\ &= x_i^T(t) (A_i^T + \bar{A}_i)\bar{x}_i(t) + \bar{x}_i^T(t) B_i x_i(t - \tau(t)) \\ &\quad + x_i^T(t) \bar{B}_i \bar{x}_i(t - \tau(t)) + \bar{x}_i^T(t) \Delta A_i(t) x_i(t) \\ &\quad + x_i^T(t) \overline{\Delta A}_i(t) \bar{x}_i(t) + \bar{x}_i^T(t) \Delta B_i(t) x_i(t - \tau(t)) \\ &\quad + x_i^T(t) \overline{\Delta B}_i(t) \bar{x}_i(t - \tau(t)) + \bar{x}_i^T(t) f_i(t, x_i(t), x_i(t - \tau(t))) \\ &\quad + x_i^T(t) \bar{f}_i(t, x_i(t), x_i(t - \tau(t))) + \bar{x}_i^T(t) K_i(x_i(t)) \\ &\quad + x_i^T(t) \bar{K}_i(x_i(t)) + \|h_i(x_i(t), x_i(t - \tau(t)))\|^2 \\ &\quad + \sum_{h=1}^N [\bar{x}_i^T(t) H_{ih}(x_i(t - \tau(t)), x_h(t - \tau(t))) \\ &\quad + x_i^T(t) \bar{H}_{ih}(x_i(t - \tau(t)), x_h(t - \tau(t)))]. \quad (20) \end{aligned}$$

From Lemma 1, we have

$$\begin{aligned} \bar{x}_i^T(t) B_i x_i(t - \tau(t)) + x_i^T(t) \bar{B}_i \bar{x}_i(t - \tau(t)) \\ \leq \|B_i\| \|x_i(t)\|^2 + \frac{1}{\|B_i\|} \|B_i x_i(t - \tau(t))\|^2 \\ \leq \|B_i\| \|x_i(t)\|^2 + \|B_i\| \|x_i(t - \tau(t))\|^2. \quad (21) \end{aligned}$$

Since $\|\Delta A_i(t)\|^2 \leq \gamma_{i1}^2$, $\|\Delta B_i(t)\|^2 \leq \gamma_{i2}^2$, combining with Lemma 1, we can obtain the following results:

$$\begin{aligned} \bar{x}_i^T(t) \Delta A_i(t) x_i(t) + x_i^T(t) \overline{\Delta A}_i(t) \bar{x}_i(t) \\ \leq \gamma_{i1} \|x_i(t)\|^2 + \frac{1}{\gamma_{i1}} \|\Delta A_i(t) x_i(t)\|^2 \leq 2\gamma_{i1} \|x_i(t)\|^2 \quad (22) \end{aligned}$$

and

$$\begin{aligned} \bar{x}_i^T(t) \Delta B_i(t) x_i(t - \tau(t)) + x_i^T(t) \overline{\Delta B}_i(t) \bar{x}_i(t - \tau(t)) \\ \leq \gamma_{i2} \|x_i(t)\|^2 + \frac{1}{\gamma_{i2}} \|\Delta B_i(t) x_i(t - \tau(t))\|^2 \\ \leq \gamma_{i2} \|x_i(t)\|^2 + \gamma_{i2} \|x_i(t - \tau(t))\|^2. \quad (23) \end{aligned}$$

According to Lemma 1 and Assumption 2, we get

$$\begin{aligned} \bar{x}_i^T(t) f_i(t, x_i(t), x_i(t - \tau(t))) \\ + x_i^T(t) \bar{f}_i(t, x_i(t), x_i(t - \tau(t))) \\ \leq L_i \|x_i(t)\|^2 + \frac{1}{L_i} \|f_i(t, x_i(t), x_i(t - \tau(t)))\|^2 \\ \leq L_i \|x_i(t)\|^2 + L_i [\|x_i(t)\|^2 + \|x_i(t - \tau(t))\|^2] \quad (24) \end{aligned}$$

and when $\Lambda_{ih} \neq 0$

$$\begin{aligned} \bar{x}_i^T(t) H_{ih}(x_i(t - \tau(t)), x_h(t - \tau(t))) \\ + x_i^T(t) \bar{H}_{ih}(x_i(t - \tau(t)), x_h(t - \tau(t))) \\ \leq \Lambda_{ih} \|x_i(t)\|^2 + \frac{1}{\Lambda_{ih}} \|H_{ih}(x_i(t - \tau(t)), x_h(t - \tau(t)))\|^2 \\ \leq \Lambda_{ih} \|x_i(t)\|^2 + \Lambda_{ih} (\|x_i(t - \tau(t))\|^2 + \|x_h(t - \tau(t))\|^2). \quad (25) \end{aligned}$$

By virtue of (18) and (21)–(25), we can rewrite (20) as

$$\begin{aligned} \mathcal{L}V_i(t, x_i(t)) \\ \leq x_i^T(t) \left[A_i^T + \bar{A}_i + \left(\|B_i\| + 2\gamma_{i1} + \gamma_{i2} + 2L_i \right. \right. \\ \left. \left. + \sum_{h=1}^N \Lambda_{ih} - k_i \right) I_n \right] \bar{x}_i(t) \\ + \left(\|B_i\| + \gamma_{i2} + L_i + \sum_{h=1}^N 2\Lambda_{ih} \right) \|x_i(t - \tau(t))\|^2 \\ + \sum_{h=1}^N \Lambda_{ih} (\|x_h(t - \tau(t))\|^2 - \|x_i(t - \tau(t))\|^2) \\ + \varpi_i \|x_i(t)\|^2 + \varsigma_i \|x_i(t - \tau(t))\|^2. \end{aligned}$$

With the notations in (19), we have

$$\begin{aligned} \mathcal{L}V_i(t, x_i(t)) \\ \leq -\sigma_i \|x_i(t)\|^2 + \xi_i \|x_i(t - \tau(t))\|^2 \\ + \sum_{h=1}^N a_{ih} F_{ih}(x_i(t - \tau(t)), x_h(t - \tau(t))) \\ = -\sigma_i V_i(t, x_i(t)) + \xi_i V_i(t - \tau(t), x_i(t - \tau(t))) \\ + \sum_{h=1}^N a_{ih} F_{ih}(x_i(t - \tau(t)), x_h(t - \tau(t))) \end{aligned}$$

where $a_{ih} = \Lambda_{ih}$ and $F_{ih}(x_i, x_h) = \|x_h\|^2 - \|x_i\|^2$.

For $t \in (s_m, t_{m+1})$, $m \in \mathbb{N}$, similarly, we have

$$\begin{aligned} \mathcal{L}V_i(t, x_i(t)) \\ = \bar{x}_i^T(t) \left[(A_i + \Delta A_i(t))x_i(t) + (B_i + \Delta B_i(t)) \right. \\ \left. \times x_i(t - \tau(t)) + \bar{f}_i(t, x_i(t), x_i(t - \tau(t))) \right. \\ \left. + \sum_{h=1}^N H_{ih}(x_i(t - \tau(t)), x_h(t - \tau(t))) \right] \\ + x_i^T(t) \left[(\bar{A}_i + \overline{\Delta A}_i(t))\bar{x}_i(t) + (\bar{B}_i + \overline{\Delta B}_i(t)) \right. \\ \left. \times \bar{x}_i(t - \tau(t)) + \bar{f}_i(t, x_i(t), x_i(t - \tau(t))) \right. \\ \left. + \sum_{h=1}^N \bar{H}_{ih}(x_i(t - \tau(t)), x_h(t - \tau(t))) \right] \\ + \|h_i(x_i(t), x_i(t - \tau(t)))\|^2 \\ \leq \lambda_{\max}(\Phi_i) V_i(t, x_i(t)) + \xi_i V_i(t - \tau(t), x_i(t - \tau(t))) \\ + \sum_{h=1}^N a_{ih} F_{ih}(x_i(t - \tau(t)), x_h(t - \tau(t))). \end{aligned}$$

Hence, condition A2 in Theorem 1 is satisfied. Moreover, by Remark 3, we have

$$\sum_{(h,i) \in \mathcal{E}(\mathcal{C}_Q)} F_{ih}(x_i, x_h) = 0.$$

That is to say, condition A3 is satisfied. By virtue of (19), (6) holds. Until now, all the conditions in Theorem 1 have been satisfied. We conclude that system (1) is globally exponentially stable in mean square. This completes the proof. ■

Remark 6: In fact, Theorem 1 is given in terms of vertex-Lyapunov functions and the topological structure of considered networks. Thus, it is of great necessity for us to find some suitable functions to verify the availability of Theorem 1, which is our aim in Theorem 2. Theorem 2 gives functions V_i and F_{ih} that satisfy conditions A1–A3 in Theorem 1. It further confirms that conditions A1–A3 are practicable and available.

Remark 7: It should be pointed out that the assumptions on functions f_i , H_{ih} , and h_i are similar to that in real-valued systems. For example, Wang *et al.* [8]–[10] put this assumption on activation functions f . The assumption on coupled functions H_{ih} can be seen in [5], [48], and [49]. And a large set of papers require disturbance strength function satisfying (17) (see [5], [14] and the references therein). Further, the assumption on control gain is general as well. If we take $K_i(x_i) = -k_i x_i$, Assumption 3 holds clearly, which is considered in [31]–[35]. What is more, we can also take $K_i(x_i) = (\tanh \|x_i\| - k_i)x_i$. Hence, the control we use has a larger application scope. Moreover, (19) is closely related to the upper bound of delay $\tau(t)$ and control interval. It is worth noting that the conditions in Theorem 2 are easily satisfied. To illustrate this point, the application and corresponding numerical examples will be given in the next section.

Remark 8: For the complex-valued networks studied in this paper, two points should be mentioned here.

- 1) Although a wide variety of interesting results have been presented for synchronization or stability of various dynamical networks via intermittent control [27]–[35], few results consider complex-valued networks.
- 2) For the research on complex-valued systems, the main difficulty is that the derivative of many functions with complex variables does not exist since they are not holomorphic. Hence, many reported results dealt with complex-valued systems by separating their real and imaginary parts [18]–[24].

In this paper, we consider the complex-valued networks directly by using the complex version Itô's formula and (conjugate) $\mathbb{R}$ derivative, which can avoid above-mentioned difficulty effectively.

C. Special Case That $s_m \rightarrow t_m$

When $s_m \rightarrow t_m$, the intermittent control works only at instants $t = t_m$. We assume that the control degenerates into

$$u_i(t) = \begin{cases} -k_i x_i(t), & t = t_m \\ 0, & t \neq t_m \end{cases} \quad m \in \mathbb{N}$$

and the intermittent controlled network is converted into the following impulsively controlled system:

$$\begin{cases} dx(t) = \left[(A_i + \Delta A_i(t))x_i(t) + (B_i + \Delta B_i(t)) \right. \\ \quad \times x_i(t - \tau(t)) + f_i(t, x_i(t), x(t - \tau(t))) \\ \quad \left. + \sum_{h=1}^N H_{ih}(x_i(t - \tau(t)), x_h(t - \tau(t))) \right] dt \\ \quad + h_i(x_i(t), x_i(t - \tau(t)))dB(t), \quad t \neq t_m \\ \Delta x_i(t) = -k_i x_i(t), \quad t = t_m, \quad m \in \mathbb{N} \end{cases} \quad (26)$$

where $\Delta x_i(t_m) = x_i(t_m^+) - x_i(t_m^-)$, $x_i(t_m^+) = \lim_{t \rightarrow t_m^+} x_i(t)$ and $x_i(t_m^-) = \lim_{t \rightarrow t_m^-} x_i(t)$. Without loss of generality, we assume that $\lim_{t \rightarrow t_m^-} x_i(t) = x_i(t_m)$, which means that the solution of system (26) is left continuous at t_m . Based on the theory of impulsive functional differential equation and comparison method, we have the following results.

Theorem 3: Suppose there exists a matrix $A = (a_{ih})_{N \times N}$ in which $a_{ih} \geq 0$ such that $(\mathcal{G}, A)$ is strongly connected and there exist positive-definite functions $V_i(t, x_i)$ ($i \in \mathcal{N}$), such that the following conditions hold.

B1: There are constants α_i , ξ_i , and η_i and functions $F_{ih}(x_i, x_h)$ such that

$$\begin{aligned} \mathcal{L}V_i(t, x_i(t)) &\leq \alpha_i V_i(t, x_i(t)) + \xi_i V_i(t - \tau(t), x_i(t - \tau(t))) \\ &\quad + \sum_{h=1}^N a_{ih} F_{ih}(x_i(t - \tau(t)), x_h(t - \tau(t))), \quad t \neq t_m \end{aligned}$$

and

$$V_i(t_m^+, x_i(t_m^+)) \leq \eta_i V_i(t_m^-, x_i(t_m^-)).$$

B2: The following inequalities hold:

$$\alpha + \frac{\ln \eta}{T} + \frac{\xi}{\eta} < 0, \quad \eta < 1$$

where $\alpha = \max_{i \in \mathcal{N}} \{\alpha_i\}$, $\xi = \max_{i \in \mathcal{N}} \{\xi_i\}$, $\eta = \max_{i \in \mathcal{N}} \{\eta_i\}$, and $T = \sup_m (t_{m+1} - t_m)$.

If conditions A1 and A3 in Theorem 1 hold, then impulsively controlled SUCVDN (26) is globally exponentially stable in mean square.

Proof: Define

$$V(t, x) = \sum_{i=1}^N c_i V_i(t, x_i)$$

where c_i is the same as that in the proof of Theorem 1. From condition A1, it is easy to get

$$\vartheta \|x\|^2 \leq V(t, x) \leq \mu \|x\|^2 \quad (27)$$

where $\mu \geq \vartheta > 0$. By condition A3, according to the proof of Theorem 1, we have

$$D^+ \mathbb{E}V(t, x(t)) \leq \alpha \mathbb{E}V(t) + \xi \mathbb{E}V(t - \tau(t), x(t - \tau(t))), \quad t \neq t_m.$$

It follows from condition B1 that:

$$\begin{aligned} V(t_m^+, x(t_m^+)) &= \sum_{i=1}^N c_i V_i(t_m^+, x_i(t_m^+)) \\ &\leq \sum_{i=1}^N c_i \eta_i V_i(t_m^-, x_i(t_m^-)) \leq \eta V(t_m^-, x(t_m^-)). \end{aligned}$$

That is $\mathbb{E}V(t_m^+, x(t_m^+)) \leq \eta \mathbb{E}V(t_m^-, x(t_m^-))$, $m \in \mathbb{N}$.

For any $\varepsilon > 0$, let $g(t)$ be a unique solution of the following impulsively delayed system:

$$\begin{cases} \dot{g}(t) = \alpha g(t) + \xi g(t - \tau(t)) + \varepsilon, & t \geq 0, t \neq t_m \\ g(t_m^+) = \eta g(t_m^-), & m \in \mathbb{N} \\ g(t) = \mu \mathbb{E} \|\phi(t)\|^2, & -\tau \leq t \leq 0. \end{cases} \quad (28)$$

Since $\mathbb{E}V(t, x(t)) \leq \mu \mathbb{E} \|\phi(t)\|^2$ for $-\tau \leq t \leq 0$, from Lemma 4, we can get $0 \leq \mathbb{E}V(t, x(t)) \leq g(t)$ for $t > 0$. Then by the formula for the variation of parameters, we get that $g(t)$ satisfies the following integral equation:

$$g(t) = \omega(t, 0)g(0) + \int_0^t \omega(t, s) [\xi g(s - \tau(s)) + \varepsilon] ds \quad (29)$$

where $\omega(t, s)$ is the Cauchy matrix of linear system

$$\begin{cases} \dot{z}(t) = \alpha z(t), & t \geq 0, t \neq t_m \\ z(t_m^+) = \eta z(t_m^-), & m \in \mathbb{N}. \end{cases}$$

Note that $0 < \eta < 1$. Then, we have

$$\begin{aligned} \omega(t, s) &= \exp(\alpha(t-s)) \prod_{s \leq t_m \leq t} \eta = \exp(\alpha(t-s)) \eta^{\frac{t-s}{T}} \\ &\leq \frac{1}{\eta} \exp(-\epsilon(t-s)) \end{aligned} \quad (30)$$

where $\epsilon = -(\alpha + (\ln \eta)/T)$. Denote $\rho =: (\mu/\eta) \sup_{-\tau \leq s \leq 0} \mathbb{E} \|\phi(s)\|^2$. It follows from (28) and (30) that:

$$\begin{aligned} g(t) &\leq \rho \exp(-\epsilon t) \\ &\quad + \int_0^t \exp(-\epsilon(t-s)) \left[\frac{\xi}{\eta} g(s - \tau(s)) + \frac{\varepsilon}{\eta} \right] ds. \end{aligned} \quad (31)$$

Note that $\alpha + (\ln \eta)/T + (\xi/\eta) < 0$. Then, we have $\epsilon - (\xi/\eta) > 0$. Let $\lambda > 0$ be the unique solution of $\lambda - \epsilon + (\xi/\eta) \exp(\lambda t) = 0$. Since $0 < \eta < 1$, for $-\tau \leq t < 0$, we have

$$\begin{aligned} g(t) &= \mu \mathbb{E} \|\phi(s)\|^2 \leq \frac{\mu}{\eta} \sup_{-\tau \leq s \leq 0} \mathbb{E} \|\phi(s)\|^2 \\ &\leq \rho \exp(-\lambda t) + \frac{\varepsilon}{\epsilon \eta - \xi}. \end{aligned} \quad (32)$$

Subsequently, we will prove (32) holds for $t \geq 0$. If it is not true, there exists a constant $\tilde{t}$ such that

$$g(t) \geq \rho \exp(-\lambda t) + \frac{\varepsilon}{\epsilon \eta - \xi}, \quad t \geq \tilde{t} \quad (33)$$

$$g(t) < \rho \exp(-\lambda t) + \frac{\varepsilon}{\epsilon \eta - \xi}, \quad t < \tilde{t}. \quad (34)$$

By (31) and (34), we have

$$\begin{aligned} g(\tilde{t}) &\leq \rho \exp(-\epsilon \tilde{t}) \\ &\quad + \int_0^{\tilde{t}} \exp(-\epsilon(\tilde{t}-s)) \left[\frac{\xi}{\eta} g(s - \tau(s)) + \frac{\varepsilon}{\eta} \right] ds \\ &< \exp(-\epsilon \tilde{t}) \left\{ \rho + \int_0^{\tilde{t}} \exp(\epsilon s) \left[\frac{\xi}{\eta} \left(\rho \exp(-\lambda(s - \tau(s))) \right. \right. \right. \\ &\quad \left. \left. \left. + \frac{\varepsilon}{\epsilon \eta - \xi} \right) + \frac{\varepsilon}{\eta} \right] ds \right\} \\ &= \exp(-\epsilon \tilde{t}) \left[\rho + \frac{\xi \rho}{\eta} \int_0^{\tilde{t}} \exp((\epsilon - \lambda)s + \lambda \tau(s)) ds \right. \\ &\quad \left. + \frac{\varepsilon \epsilon}{\eta \epsilon - \xi} \int_0^{\tilde{t}} \exp(\epsilon s) ds \right] \end{aligned}$$

$$\begin{aligned} &\leq \exp(-\epsilon \tilde{t}) \left[\rho + \frac{\rho \xi \exp(\lambda \tau)}{\eta(\epsilon - \lambda)} \exp((\epsilon - \lambda)\tilde{t}) \right. \\ &\quad \left. - \frac{\rho \xi}{\eta(\epsilon - \lambda)} \exp(\lambda \tau) + \frac{\varepsilon}{\eta \epsilon - \xi} (\exp(\epsilon \tilde{t}) - 1) \right] \\ &< \rho \exp(-\lambda \tilde{t}) + \frac{\varepsilon}{\epsilon \eta - \xi} \end{aligned}$$

which contradicts (33). Therefore, (32) holds for $t \geq 0$. Letting $\varepsilon \rightarrow 0$, it follows from (32) that:

$$\mathbb{E}V(t, x(t)) \leq g(t) \leq \rho \exp(-\lambda t).$$

Then by (27), it yields $\mathbb{E} \|x(t)\|^2 \leq (\rho/\vartheta) \exp(-\lambda t) = (\mu/\eta\theta) \sup_{-\tau \leq s \leq 0} \mathbb{E} \|\phi(s)\|^2 \exp(-\lambda t)$. Therefore, by Definition 1, it is easy to see that impulsively controlled SUCVDN (26) is globally exponentially stable in mean square. This completes the proof. ■

Then, based on Theorem 3, another easily verified theorem is given as follows.

Theorem 4: Suppose that Assumption 2 holds and $(\mathcal{G}, (\Lambda_{ih})_{N \times N})$ is strongly connected. If there exists η_i , such that $(1 - k_i)^2 \leq \eta_i$, $\eta = \max_{i \in \mathcal{N}} \{\eta_i\} < 1$ and $\alpha + (\ln \eta)/T + (\xi/\eta) < 0$, where α , ξ , and T are defined in Theorem 2, then impulsively controlled system (26) is globally exponentially stable in mean square.

Let $V_i(t, x_i) = x_i^T \tilde{x}_i$. The rest is similar to the proof of Theorem 2 so we omit it.

Remark 9: It should be noticed that the previous results [27]–[35] on intermittent control did not consider the special case that control width tends to zero. However, considering the practical meaning of impulsive control, in this paper, we also consider the stabilization of SUCVDNs via impulsive control.

IV. APPLICATION TO STOCHASTIC COUPLED OSCILLATORS

Coupled oscillators, which come from problems in applied science, have been a heated and interesting topic. Due to their widespread applications to describing various physical and biological phenomena. And great emphasis has been attached to them. Recently, complex-valued oscillators have become an important topic as well [51]. Therefore, in this section, to check the correctness of our results above, we apply the main results to discussing the stabilization problem of stochastic complex-valued coupled oscillators with time-varying delay which is described on digraph $\mathcal{G}$ as follows:

$$\begin{aligned} \ddot{x}_i(t) &+ \varphi_i \dot{x}_i(t) + \alpha_i x_i(t - \tau(t)) + f_i(t, x_i(t), x_i(t - \tau(t))) \\ &\quad + \sum_{h=1}^N a_{ih} (x_h(t - \tau(t)) - x_i(t - \tau(t))) \\ &= \tilde{h}_i(x_i(t), x_i(t - \tau(t))) \dot{B}(t), \quad i \in \mathcal{N} \end{aligned} \quad (35)$$

where $\varphi_i \in \mathbb{C}^1$, $f_i : \mathbb{R}_+ \times \mathbb{C}^1 \times \mathbb{C}^1 \rightarrow \mathbb{C}^1$ is a continuously vector-valued function. Also, we assume that the intensity of influence from vertex i to vertex h is described by $a_{ih} \geq 0$ and $a_{ih} = 0$ if and only if there exists no arc from i to h in the graph $\mathcal{G}$. This kind of real-valued coupled oscillators has been widely studied in [5], [12], [44], and [48]–[50].

We introduce $y_i(t) = \dot{x}_i(t) + \lambda_i x_i(t)$, where λ_i are suitable complex-valued constants. Hence, we have

$$\begin{cases} dx_i(t) = [y_i(t) - \lambda_i x_i(t)] dt \\ dy_i(t) = \left[(-\varphi_i + \lambda_i) y_i(t) + \lambda_i (\varphi_i - \lambda_i) x_i(t) \right. \\ \quad - \alpha_i x_i(t - \tau(t)) - f_i(t, x_i(t), x_i(t - \tau(t))) \\ \quad - \sum_{h=1}^N a_{ih}(x_h(t - \tau(t)) - x_i(t - \tau(t))) \\ \quad \left. + \tilde{h}_i(x_i(t), x_i(t - \tau(t))) dB(t), \quad i \in \mathcal{N}. \right] \end{cases} \quad (36)$$

For the sake of convenience, let $X_i = (x_i, y_i)^T$, $\tilde{X}_i = (\tilde{x}_i, \tilde{y}_i)^T$, $A_i = \begin{bmatrix} -\lambda_i & 1 \\ \lambda_i(\varphi_i - \lambda_i) & -\varphi_i + \lambda_i \end{bmatrix}$, $B_i = \begin{bmatrix} 0 & 0 \\ -\alpha_i & 0 \end{bmatrix}$

$$F_i(t, X_i, \tilde{X}_i) = (0, -f_i(t, x_i, \tilde{x}_i))^T \quad (37)$$

$$H_{ih}(X_i, X_h) = (0, -a_{ih}(x_h - x_i))^T \quad (38)$$

$$h_i(X_i, \tilde{X}_i) = (0, \tilde{h}_i(x_i, \tilde{x}_i))^T. \quad (39)$$

Considering parameter uncertainties, we let $\varphi_i \rightarrow \varphi_i + \Delta\varphi_i(t)$, $\alpha_i \rightarrow \alpha_i + \Delta\alpha_i(t)$. Thus, we have

$$\Delta A_i(t) = \begin{bmatrix} 0 & 0 \\ \lambda_i \Delta\varphi_i(t) & -\Delta\varphi_i(t) \end{bmatrix}, \Delta B_i(t) = \begin{bmatrix} 0 & 0 \\ -\Delta\alpha_i(t) & 0 \end{bmatrix}. \quad (40)$$

So (36) with uncertain parameters can be rewritten as

$$\begin{aligned} dX_i(t) = & \left[(A_i + \Delta A_i(t))X_i(t) + (B_i + \Delta B_i(t))X_i(t - \tau(t)) \right. \\ & + F_i(t, X_i(t), X_i(t - \tau(t))) \\ & + \sum_{h=1}^N H_{ih}(X_i(t - \tau(t)), X_h(t - \tau(t))) \\ & \left. + h_i(X_i(t), X_i(t - \tau(t))) dB(t), \quad i \in \mathcal{N}. \right] \end{aligned} \quad (41)$$

Taking control into consideration, we describe system (41) as

$$\begin{aligned} dX_i(t) = & \left[(A_i + \Delta A_i(t))X_i(t) + (B_i + \Delta B_i(t))X_i(t - \tau(t)) \right. \\ & + F_i(t, X_i(t), X_i(t - \tau(t))) \\ & + \sum_{h=1}^N H_{ih}(X_i(t - \tau(t)), X_h(t - \tau(t))) + U_i(t) \\ & \left. + h_i(X_i(t), X_i(t - \tau(t))) dB(t), \quad i \in \mathcal{N} \right] \end{aligned} \quad (42)$$

where $U_i(t)$ is the control input designed as

$$U_i(t) = \begin{cases} K_i(X_i(t)), & t_m \leq t \leq s_m \\ 0, & s_m < t < t_{m+1} \end{cases} \quad m \in \mathbb{N}.$$

We next give some necessary assumptions.

D1: There exist positive constants L_i , ϖ_i , and ς_i such that

$$\|f_i(t, x_i, \tilde{x}_i)\|^2 \leq L_i^2 (\|x_i\|^2 + \|\tilde{x}_i\|^2) \quad (43)$$

and

$$\|\tilde{h}_i(x_i, \tilde{x}_i)\|^2 \leq \varpi_i \|x_i\|^2 + \varsigma_i \|\tilde{x}_i\|^2. \quad (44)$$

D2: There is positive constant k_i such that

$$\tilde{X}_i^T K_i(X_i) + X_i^T \tilde{K}_i(X_i) \leq -k_i \|X_i\|^2.$$

D3: For all $t \geq 0$, there exist positive constants γ_{i1} and γ_{i2} such that

$$\max \left\{ \|\lambda_i \Delta\varphi_i(t)\|^2, \|\Delta\varphi_i(t)\|^2 \right\} \leq \gamma_{i1}^2 \\ \|\Delta\alpha_i(t)\|^2 \leq \gamma_{i2}^2.$$

Theorem 5: Suppose that Assumption 1 is satisfied and conditions D1–D3 hold. Let $(\mathcal{G}, (a_{ih})_{N \times N})$ be strongly connected. If the following inequalities hold:

$$(\theta - \tau)\lambda - (T - \theta)(\alpha + \xi) > 0, \quad \sigma > \xi \quad (45)$$

where $\alpha = \max_{i \in \mathcal{N}} \{\lambda_{\max}(\Phi_i)\}$, $\sigma = \min_{i \in \mathcal{N}} \{\sigma_i\}$, $\xi = \max_{i \in \mathcal{N}} \{\xi_i\}$, $\sigma_i = \lambda_{\min}(k_i I_2 - \Phi_i) > 0$, $\xi_i = \|B_i\| + \gamma_{i2} + 2L_i + 2 \sum_{h=1}^N a_{ih} + \varsigma_i$, $\Phi_i = \bar{A}_i + A_i^T + (\|B_i\| + 2\gamma_{i1} + \gamma_{i2} + 2L_i + 2 \sum_{h=1}^N a_{ih} + \varpi_i) I_2$, $\lambda > 0$ is the real root of equation $\sigma - \lambda - \xi \exp(\lambda\tau) = 0$, then system (42) is globally exponentially stable in mean square.

Proof: It follows from (40) and condition D3 that:

$$\begin{aligned} (\Delta A_i(t))^T \overline{\Delta A_i(t)} &= \begin{bmatrix} \|\lambda_i \Delta\varphi_i(t)\|^2 & 0 \\ 0 & \|\Delta\varphi_i(t)\|^2 \end{bmatrix} \leq \gamma_{i1}^2 I_2 \\ (\Delta B_i(t))^T \overline{\Delta B_i(t)} &= \begin{bmatrix} \|\Delta\alpha_i(t)\|^2 & 0 \\ 0 & 0 \end{bmatrix} \leq \gamma_{i2}^2 I_2. \end{aligned}$$

This implies that $\|\Delta A_i(t)\|^2 \leq \gamma_{i1}^2$ and $\|\Delta B_i(t)\|^2 \leq \gamma_{i2}^2$. Then by (37) and (43), we can obtain

$$\begin{aligned} \|F_i(t, X_i, \tilde{X}_i)\|^2 &= \|f_i(t, x_i, \tilde{x}_i)\|^2 \\ &\leq L_i^2 (\|x_i\|^2 + \|\tilde{x}_i\|^2) \\ &\leq L_i^2 (\|X_i\|^2 + \|\tilde{X}_i\|^2). \end{aligned}$$

From (38), the following inequality is satisfied:

$$\begin{aligned} \|H_{ih}(X_i, X_h)\|^2 &= \|-a_{ih}(x_h - x_i)\|^2 \\ &\leq 2a_{ih} (\|x_h\|^2 + \|x_i\|^2) \\ &\leq 2a_{ih} (\|X_h\|^2 + \|X_i\|^2). \end{aligned}$$

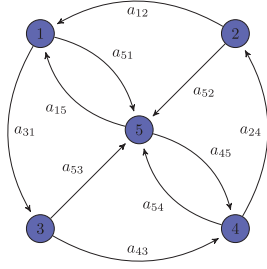
Similarly, from (39) and (44), we derive that

$$\begin{aligned} \|h_i(X_i, \tilde{X}_i)\|^2 &= \|\tilde{h}_i(x_i, \tilde{x}_i)\|^2 \\ &\leq \varpi_i \|x_i\|^2 + \varsigma_i \|\tilde{x}_i\|^2 \\ &\leq \varpi_i \|X_i\|^2 + \varsigma_i \|\tilde{X}_i\|^2. \end{aligned}$$

It is clear that Assumption 2 is satisfied. Moreover, from condition D2, it is clear that Assumption 3 holds. Based on (45), we can derive that (19) in Theorem 2 holds. Thus, we can check that all the conditions of Theorem 2 have been fulfilled. By virtue of Theorem 2, we obtain that system (42) is globally exponentially stable in mean square. This completes the proof. ■

When $s_m \rightarrow t_m$, according to Section III-C, system (42) is converted into the following impulsively system:

$$\begin{cases} \dot{X}(t) = \left[(A_i + \Delta A_i(t))X_i(t) + (B_i + \Delta B_i(t)) \right. \\ \quad \times X_i(t - \tau(t)) + F_i(t, X_i(t), X(t - \tau(t))) \\ \quad \left. + \sum_{h=1}^N H_{ih}(X_i(t - \tau(t)), X_h(t - \tau(t))) \right] dt \\ \quad + h_i(X_i(t), X_i(t - \tau(t))) dB(t), \quad t \neq t_m \\ \Delta X_i(t) = -k_i X_i(t), \quad t = t_m, \quad m \in \mathbb{N}. \end{cases} \quad (46)$$

Fig. 1. Strongly connected digraph $\mathcal{G}$ with five vertices.

Then, based on Theorem 4, we give another theorem which can be checked easily.

Theorem 6: Let conditions D1–D3 hold and $(\mathcal{G}, (a_{ih})_{N \times N})$ be strongly connected. If there exists η_i , such that $(1 - k_i)^2 \leq \eta_i$ and $\eta = \max_{i \in \mathcal{N}} \{\eta_i\}$ satisfy $\eta < 1$ and $\alpha + (\ln \eta / T) + (\xi / \eta) < 0$, where α , ξ , and T are defined in Theorem 5, then impulsively controlled system (46) is globally exponentially stable in mean square.

The proof process of Theorem 6 is similar to that of Theorem 5, so we omit the proof of Theorem 6.

Remark 10: Recently, coupled oscillators has been widely studied since their applications in engineering and many results on coupled oscillators have been reported [5], [12], [44], [48]–[50]. However, these results are all based on real-valued systems. From [51], we know that complex-valued oscillators have some practical applications as well. And to the best of authors' knowledge, there are few results on complex-valued coupled oscillators. Hence, this paper has important leading significance for further study of complex-valued coupled oscillators.

V. NUMERICAL EXAMPLES

In this section, two numerical examples, which are related to the stochastic coupled oscillators studied in Section IV, are given to illustrate the effectiveness and feasibility of the obtained results.

Example 1: Consider stochastic coupled oscillators (41) on a digraph $(\mathcal{G}, A)$ with five vertices shown in Fig. 1 and the weight matrix is assumed as

$$A = \begin{bmatrix} 0 & 0.1 & 0 & 0 & 0.05 \\ 0 & 0 & 0 & 0.15 & 0 \\ 0.15 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0.1 & 0 & 0.05 \\ 0.05 & 0.05 & 0.05 & 0.05 & 0 \end{bmatrix}.$$

It is obvious that the digraph $(\mathcal{G}, A)$ is strongly connected. We select $\tau(t) = 0.1 \sin^2 t$ and functions $f_i(t, x_i, \tilde{x}_i)$, $\tilde{h}_i(x_i, \tilde{x}_i)$, $\Delta \alpha_i(t)$, and $\Delta \varphi_i(t)$, parameters λ_i , φ_i , and α_i , $i = 1, 2, \dots, 5$, as follows:

$$f_i(x_i, y_i) = 0.3(\tanh \|x_i\|x_i + \tanh \|y_i\|y_i), \quad i = 1, 2, \dots, 5$$

$$\tilde{h}_i(x_i, y_i) = \begin{cases} 0.4x_1 + 0.3y_1, & i = 1, 2 \\ 0.3x_2 + (0.2 + 0.2i)y_2 \\ (0.2 + 0.3i)x_3 + (0.1 + 0.3i)y_3 \\ (0.3 + 0.1i)x_4 + (0.1 + 0.3i)y_4 \\ (0.2 + 0.2i)x_5 + (0.2 + 0.2i)y_5 \end{cases}$$

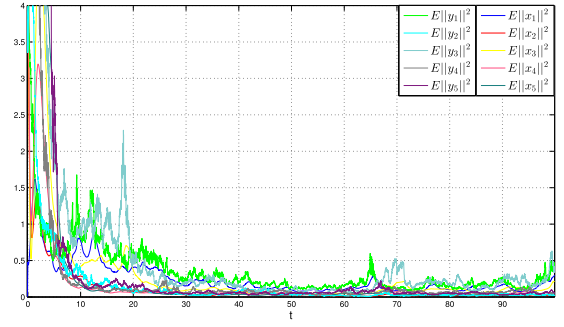


Fig. 2. Mean square trajectory of system (41) without control.

TABLE I
VALUES OF PARAMETERS ϖ_i , ς_i , k_i , $\lambda_{\max}(\Phi_i)$, σ_i , AND ξ_i

i	ϖ_i	ς_i	k_i	$\lambda_{\max}(\Phi_i)$	σ_i	ξ_i
1	0.32	0.18	3.6	2.0373	4.7204	1.01
2	0.18	0.16	3.4	2.3463	4.677	0.94
3	0.26	0.2	3.6	2.5597	4.556	0.98
4	0.2	0.2	3.6	2.6893	4.7284	1
5	0.16	0.16	3.8	2.6267	4.7359	0.96

$\Delta \alpha_i(t) = \Delta \varphi_i(t) = 0.2 \sin^2 t$, $\lambda_1 = \lambda_2 = 0.9 + 0.4i$, $\lambda_3 = \lambda_4 = 1.1 + 0.5i$, $\lambda_5 = 1.1 + 0.5i$, $\varphi_1 = \varphi_2 = 1.2 + 0.5i$, $\varphi_3 = \varphi_4 = 1.2 + 0.4i$, $\varphi_5 = 1.1 + 0.6i$, $\alpha_1 = 0.2 - 0.3i$, $\alpha_2 = \alpha_3 = 0.2 - 0.2i$, and $\alpha_4 = \alpha_5 = 0.1 + 0.3i$.

Then we take initial conditions as $x_1(t) = (1 + 2i) \sin(\pi t)$, $y_1(t) = 2 \cos(\pi t) - 3 \sin(\pi t)i$, $x_2(t) = 3 \sin(\pi t) + 2 \sin(\pi t)i$, $y_2(t) = -22 \cos^2(\pi t) - 3 \sin^2(\pi t)i$, $x_3(t) = 4 \sin(2\pi t) - 2 \cos(2\pi t)i$, $y_3(t) = -3 \sin(\pi t) \cos(2\pi t) + 4 \cos(2\pi t)i$, $x_4(t) = 1 + \cos(\pi t) - 2i$, $y_4 = -2 + \cos(3\pi t)i$, $x_5(t) = \cos^2(3\pi t) + 3i$, $y_5 = (-3 + 5i) \cos(\pi t)$, $t \in [-0.1, 0]$. The corresponding simulation results for system (41) are shown in Fig. 2. From Fig. 2, we can see that system (41) is unstable.

In order to stabilize system (41), we take

$$K_i(X_i) = \begin{bmatrix} \tanh(x_i^R) + i \tanh(x_i^I) - \tilde{K}_i x_i \\ \tanh(y_i^R) + i \tanh(y_i^I) - \tilde{K}_i y_i \end{bmatrix}$$

where $\tilde{K}_i = 2.8$ for $i = 1, 3, 4$, $\tilde{K}_2 = 2.7$ and $\tilde{K}_5 = 2.9$. And that the intermittent control exists on time span

$$[0, 0.65] \cup [0.95, 1.58] \cup [1.91, 2.52] \cup [2.89, 3.49] \\ \cup [3.88, 4.48] \cup [4.88, 5.48] \dots$$

Then we get $\theta = 0.6$, $T = 1$.

By calculations, we choose $L_i = 0.18$, $\gamma_{i1}^2 = 0.0464$, and $\gamma_{i2}^2 = 0.04$ and parameters ϖ_i , ς_i , k_i , $\lambda_{\max}(\Phi_i)$, σ_i , and ξ_i in Table I.

According to these chosen parameters, we can get that conditions D1–D3 hold. Moreover, from Table I, we have $\alpha = \max_{i \in \{1, \dots, 5\}} \{\lambda_{\max}(\Phi_i)\} = 2.6893$, $\sigma = \min_{i \in \{1, \dots, 5\}} \{\sigma_i\} = 4.556$, $\xi = \max_{i \in \{1, \dots, 5\}} \{\xi_i\} = 1.01$. It is obvious that $\sigma > \xi$. Further, according to the definition of λ [i.e., $\lambda > 0$ is the unique real root of equation $\sigma - \lambda - \xi \exp(\lambda \tau) = 0$], we get $\lambda = 3.1694$. Then we have $(\theta - \tau)\lambda - (T - \theta)(\alpha + \xi) = 0.105 > 0$. Until now, all conditions of Theorem 5 are satisfied, which implies that system (42) is globally exponentially stable in mean square. Taking the same initial conditions as above, we give the simulation results of system (42) in Fig. 3.

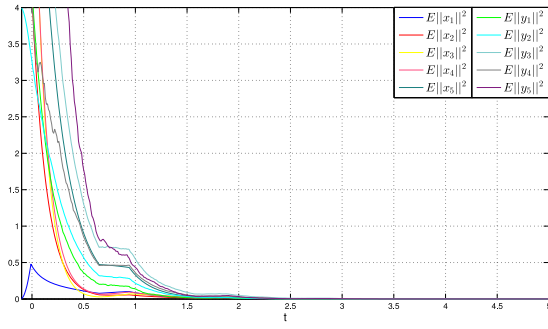


Fig. 3. Mean square trajectory of system (42) with intermittent control.

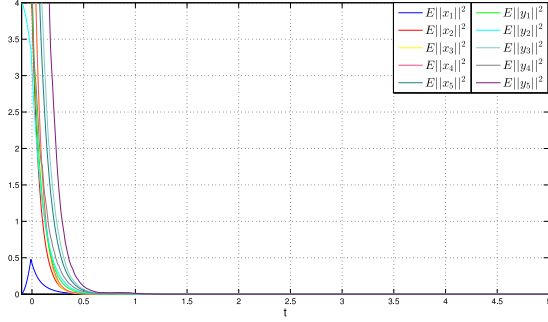


Fig. 4. Mean square trajectory of system (42) with intermittent control that has larger control gain.

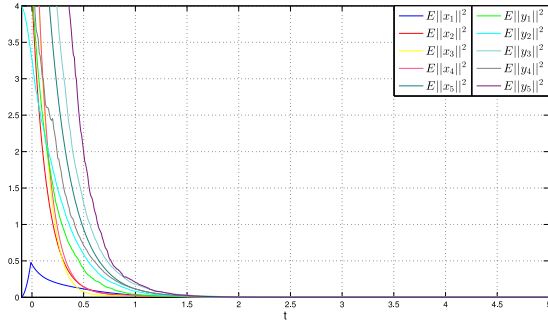


Fig. 5. Mean square trajectory of system (42) with continuous control.

We can see that system (42) is stable, which manifests the effectiveness of the proposed results.

On the other hand, if we let $K_i(X_i) = -5X_i$, $i = 1, 2, \dots, 5$ when intermittent control works, and keep the same values of other parameters as above, it is easy to get that all conditions of Theorem 5 are still satisfied. And control $K_i(X_i)$ makes k_i larger, i.e., $K_i(X_i)$ has larger control gain. Taking the same initial conditions, we present the simulation results in Fig. 4. Compared with Fig. 3, we can see that larger control gain makes system have better stability. Moreover, if we let the intermittent control work all the time, this intermittent control becomes common feedback control. So we have $\theta = 1$. Obviously, all conditions of Theorem 5 still hold. Then we present the simulation results in Fig. 5. Compared with Fig. 3, we can see that larger θ makes better stability. These simulation results all illustrate the effectiveness of our theoretical results.

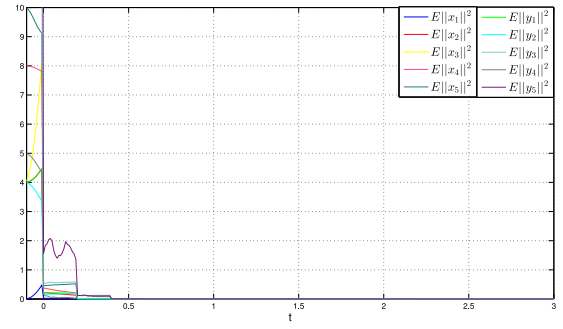


Fig. 6. Mean square trajectory of impulsively controlled system (46).

Example 2: Consider impulsively system (46) with the same parameters in Example 1 except t_m and K_i . According to Example 1, we have that conditions D1–D3 hold. In this example, we choose $t_m = 0.2$ m, $m \in \mathbb{N}$, and $K_i(X_i) = -k_i X_i$, where $k_i = 0.8$ for $i = 1, 2$ and $k_i = 1.2$ for $i = 3, 4, 5$. Hence, we have $T = 0.2$ and $\eta_i = 0.2$, which implies that $\eta = 0.2$. Then by calculation, we get $\alpha + (\ln \eta / T) + (\xi / \eta) = -0.308 < 0$. Hence, all conditions in Theorem 6 are fulfilled. Thus, system (46) is globally exponentially stable in mean square. Take the same initial conditions in Example 1, we have simulation results of impulsively controlled system (46) in Fig. 6. From Fig. 6, we can see that the simulation results have a good agreement with the theoretical analysis obtained in this paper.

VI. CONCLUSION

In this paper, we investigated the stabilization problem of SUCVDNs via aperiodically intermittent nonlinear control and impulse control, respectively. By introducing complex version Itô's formula and (conjugate) $\mathbb{R}$ derivative, we studied SUCVDNs directly. Then integrating the Lyapunov method and graph theory, we obtained some new stability criteria for SUCVDNs. Based on this paper, the following interesting problems will be studied in the future.

- 1) State estimation has become an important issue in the past several decades [52], [53], since system states are always hard to be measured. Designing an estimator for complex-valued networks will be our future work.
- 2) Sliding model control provides a powerful tool to deal with systems with uncertainty and has the advantages of quick response and strong robustness [8]–[10]. We will consider uncertain complex-valued systems via sliding model control.
- 3) Since Markovian jump systems provide a powerful way to model many practical systems whose structure is subject to random abrupt change [39]–[41]. Thus, complex-valued systems with Markovian switching will be our next work as well.
- 4) The real network model may suffer from intermittent measurements, uncertain sensing, or unreliable communication. These limited capacities over transiting process can well be modeled by multiplicative noises [54]–[56], which will be our future work.

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